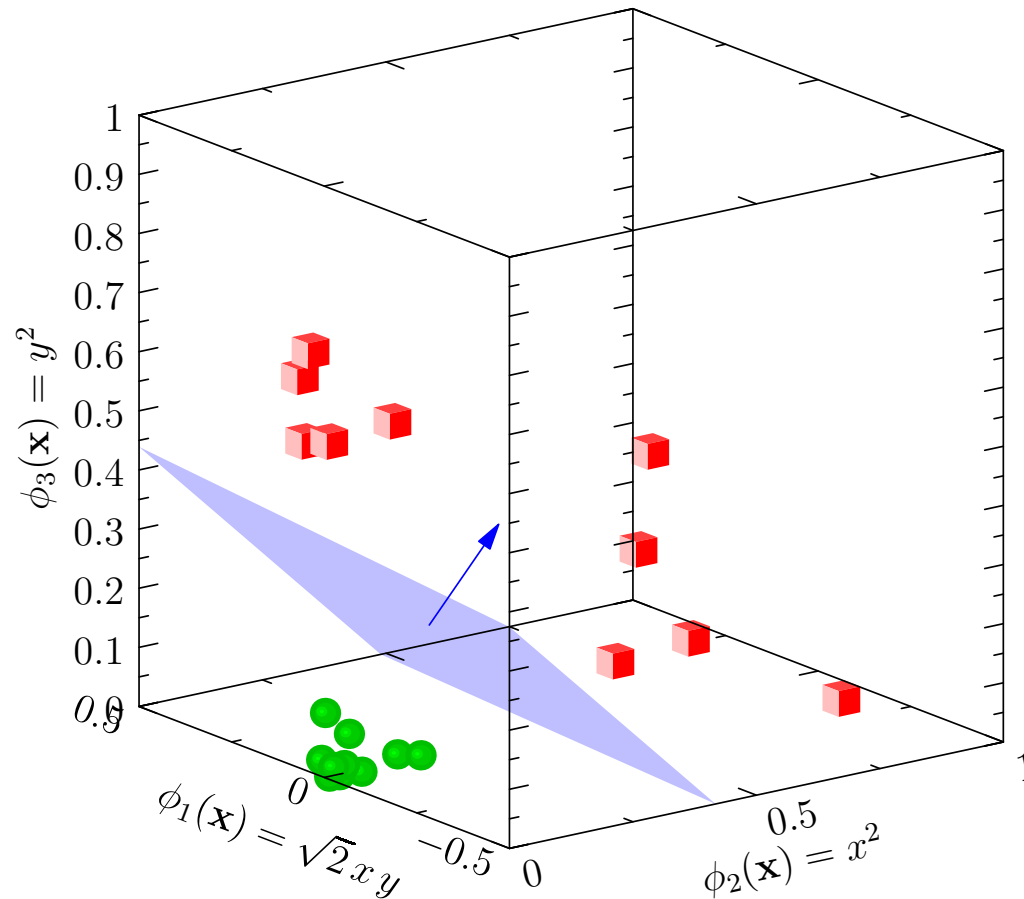


Mathematics of Machine Learning

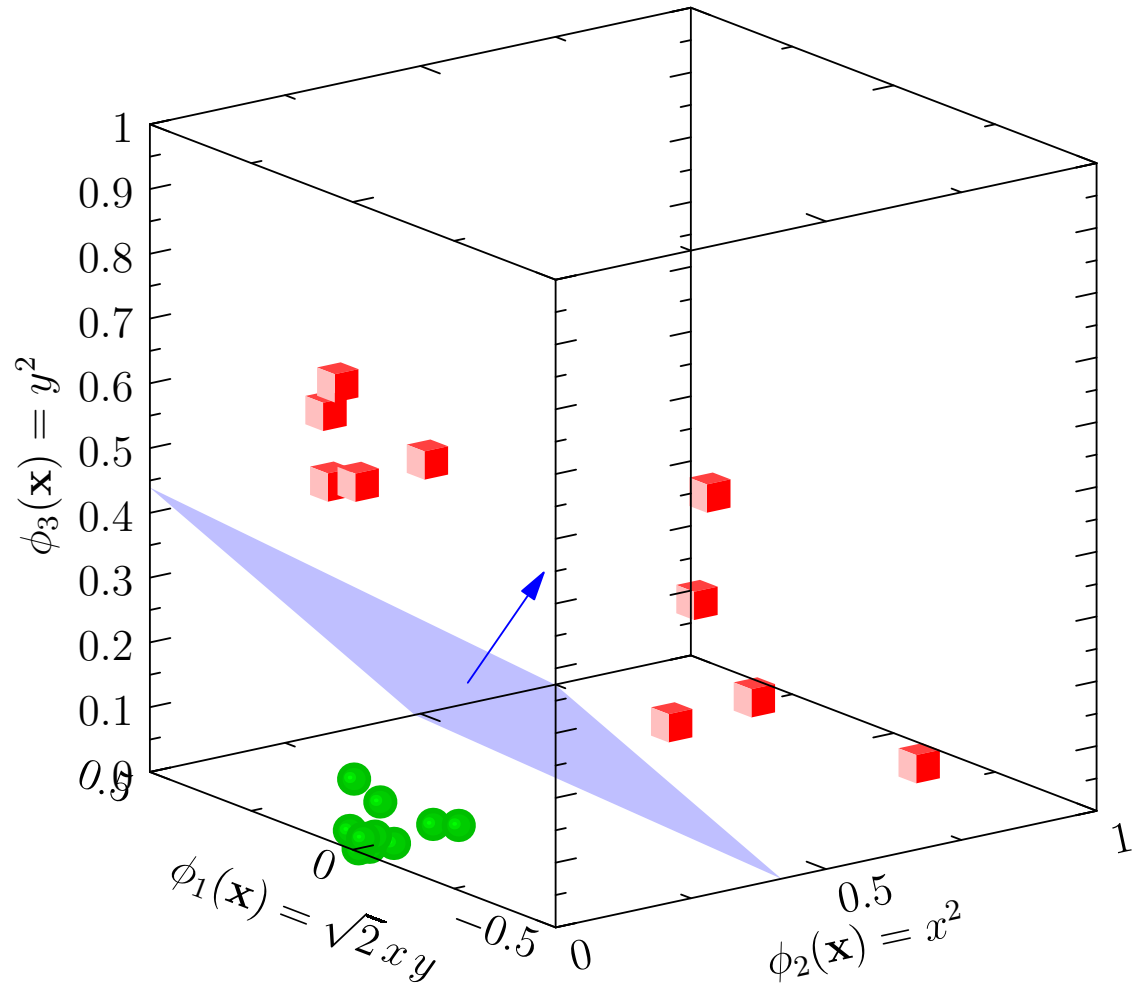
Kernel Trick



The Kernel Trick, SVMs, Regression

Outline

1. The Kernel Trick
2. Positive Semi-Definite Kernels
3. Kernel Properties
4. Beyond Classification



SVM Kernels

- SVM Kernels are functions of two variables that can be factorised

$$K(\mathbf{x}, \mathbf{y}) = \langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle = \sum_i \phi_i(\mathbf{x}) \phi_i(\mathbf{y})$$

- where $\phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots)^\top$ and $\phi_i(\mathbf{x})$ are real valued functions of $\mathbf{x}$
- $K(\mathbf{x}, \mathbf{y})$ will be positive semi-definite (because it is an inner-product)
- Furthermore, any positive semi-definite function will factorise
- This factorisation is not always obvious (we return to this later)

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Dual Form

- Recall that the dual problem for an SVM is

$$\max_{\alpha} \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$$

- subject to $\sum_{k=1}^m y_k \alpha_k = 0$ and $0 \leq \alpha_k (\leq C)$
- But since $K(\mathbf{x}_k, \mathbf{x}_l) = \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$ the dual problem becomes

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- This is the **kernel trick**

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- This is the **kernel trick**—we never have to compute $\phi(\mathbf{x})$!

Classifying New Data

- Having trained the SVM we now have to use it
- Given a new input $\mathbf{x}$ we decide on the class

$$y = \text{sgn}(\langle \mathbf{w}, \phi(\mathbf{x}) \rangle - b) \quad \text{but} \quad \mathbf{w} = \sum_{k=1}^m \alpha_k y_k \phi(\mathbf{x}_k)$$

- In the dual representation this becomes

$$\text{sgn} \left(\sum_{k=1}^m \alpha_k y_k K(\mathbf{x}_k, \mathbf{x}) - b \right)$$

where we only need to sum over the non-zero α_k (i.e. the support vectors SVs)

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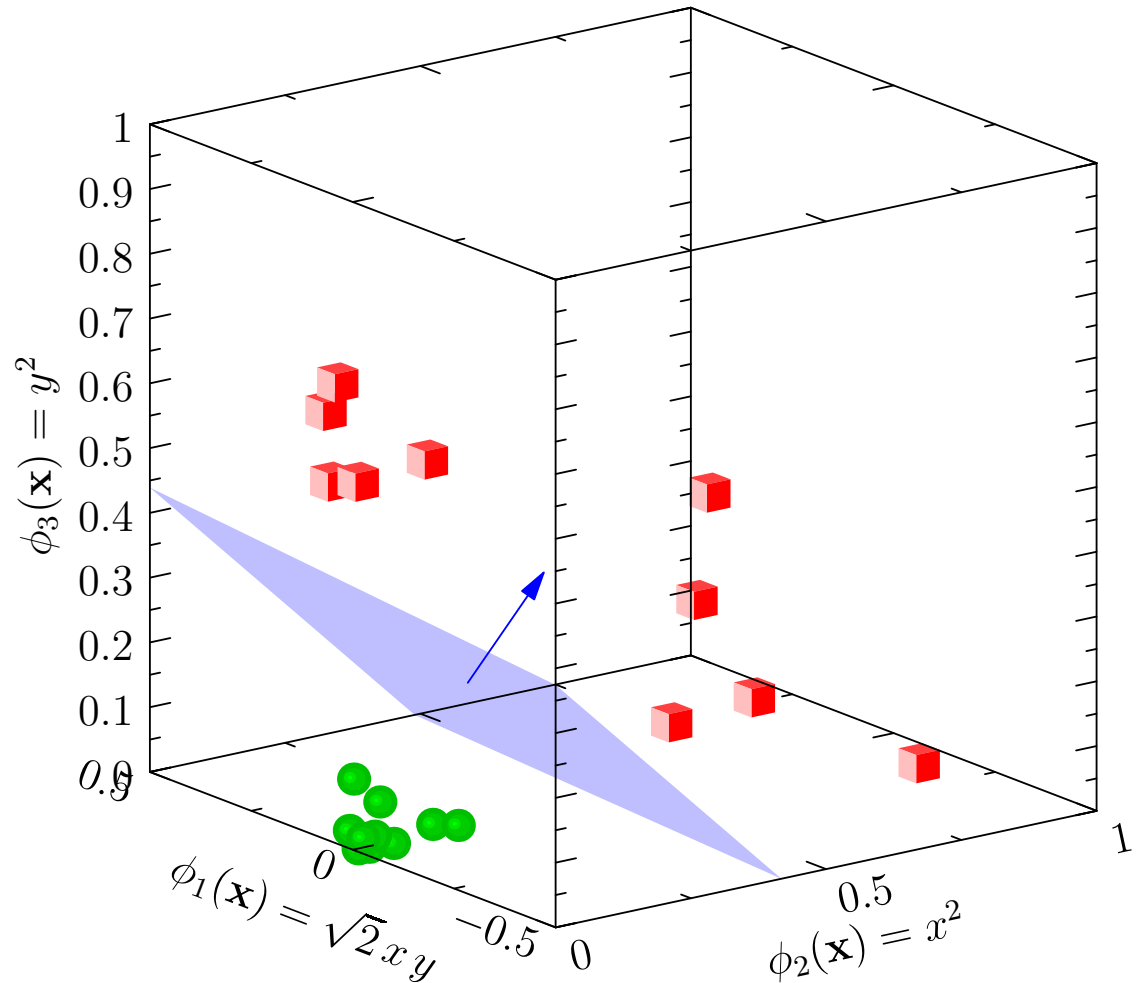
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Recap on Eigen Systems

- Recall for a symmetric $(n \times n)$ matrix $\mathbf{M}$ an eigenvector, $\mathbf{v}$

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$$

- There are n independent eigenvectors $\mathbf{v}^{(i)}$ with real eigenvalues $\lambda^{(i)}$
- The eigenvectors are orthogonal so that $\mathbf{v}^{(i)\top}\mathbf{v}^{(j)} = 0$ if $i \neq j$
- Forming a matrix of eigenvectors $\mathbf{V} = (\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(n)})$ the matrix satisfies

$$\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}$$

- Such matrices are said to be orthogonal

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Eigen Decomposition

- From the eigenvalue equation $\mathbf{M}\mathbf{v}^{(k)} = \lambda^{(k)}\mathbf{v}^{(k)}$

$$\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{\Lambda} \quad \text{where} \quad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Multiplying on the right by $\mathbf{V}^T$ we get

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \sum_{k=1}^n \lambda^{(k)} \mathbf{v}^{(k)} \mathbf{v}^{(k)T}$$

Or

$$M_{ij} = \sum_{k=1}^n \lambda^{(k)} v_i^{(k)} v_j^{(k)}$$

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$$M_{ij} = \sum_{k=1}^n \lambda^{(k)} v_i^{(k)} v_j^{(k)} = \sum_{k=1}^n u_i^{(k)} u_j^{(k)} = \langle \mathbf{u}_i, \mathbf{u}_j \rangle$$

$$u_i^{(k)} = \sqrt{\lambda^{(k)}} v_i^{(k)}$$

Eigenfunctions

- By analogy for a symmetric function of two variables we can define an *eigenfunction*

$$\int K(\mathbf{x}, \mathbf{y}) \psi(\mathbf{y}) d\mathbf{y} = \lambda \psi(\mathbf{x})$$

- In general there will be a denumerable set of eigenfunctions $\psi^{(k)}(\mathbf{x})$ where

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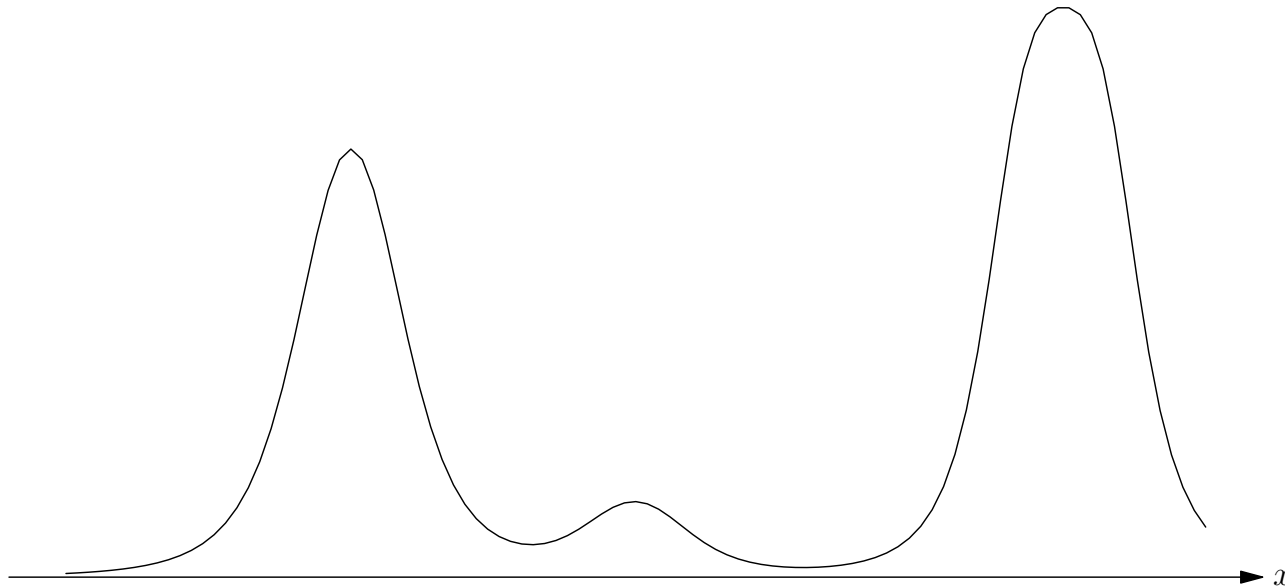
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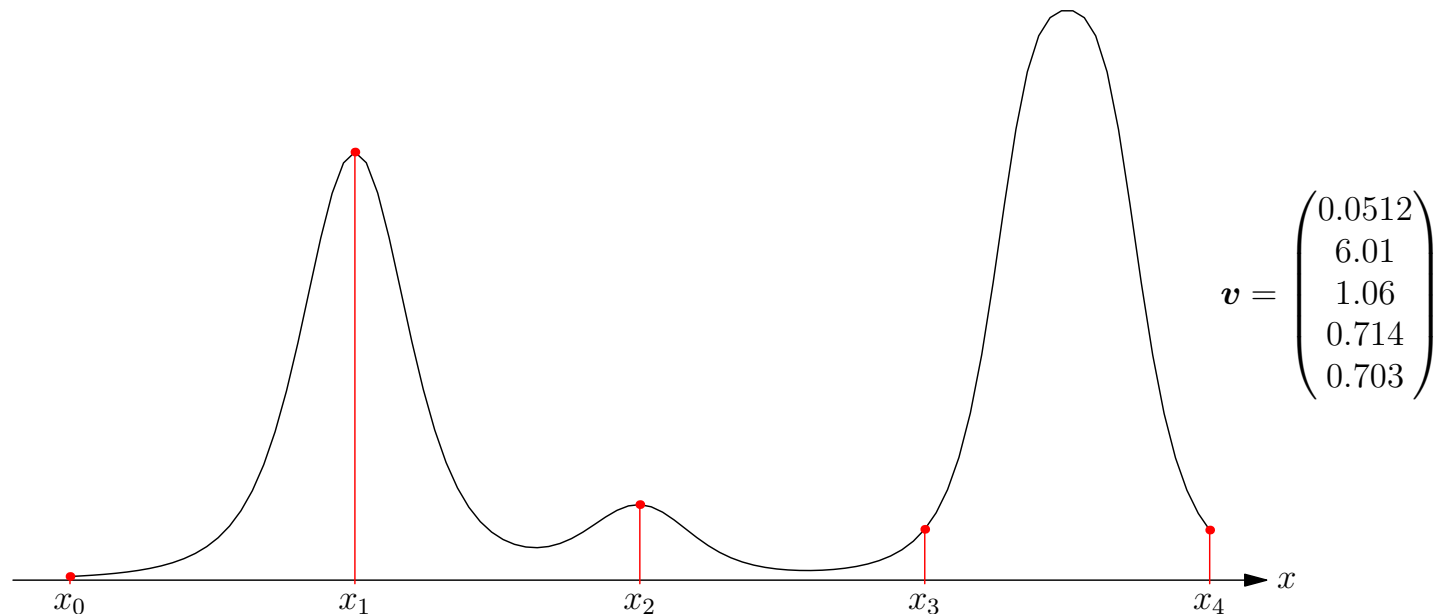
- Consider sampling a function at a set of points



- In the limit where the number of sample points goes to infinity the vector more closely approximates a function
- The k^{th} component of the vector corresponds to $f(x_k)$

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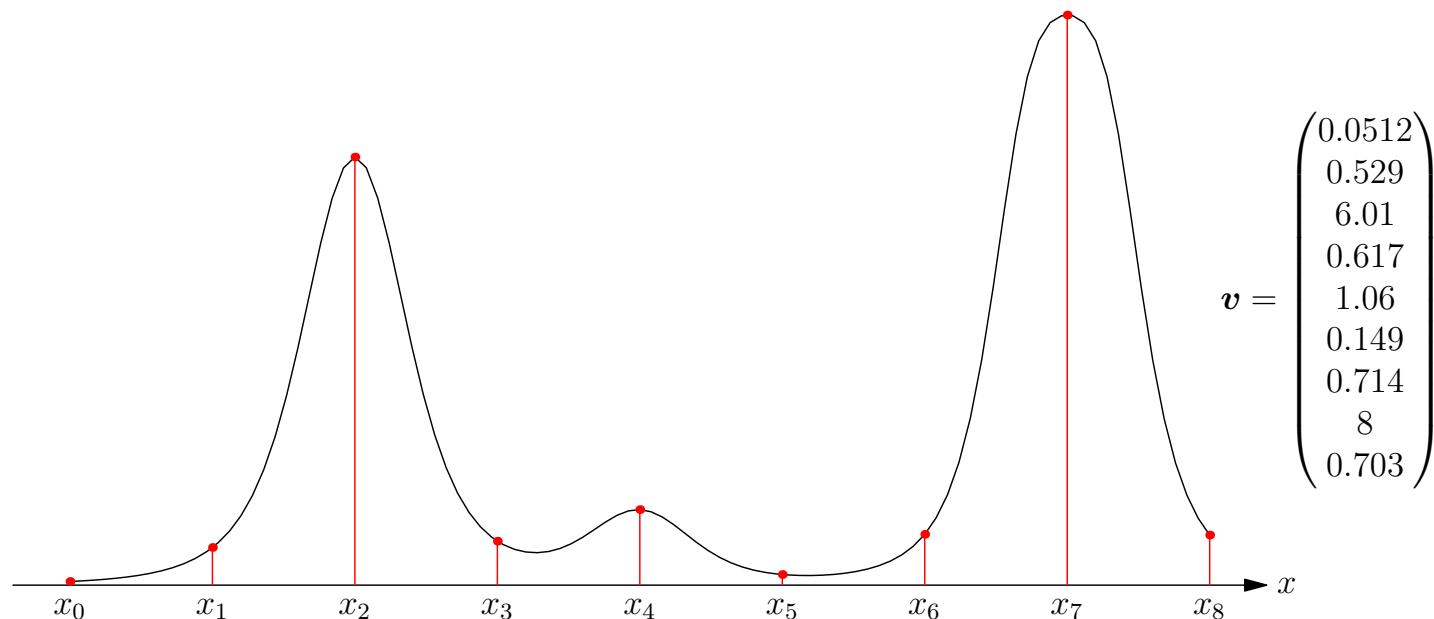
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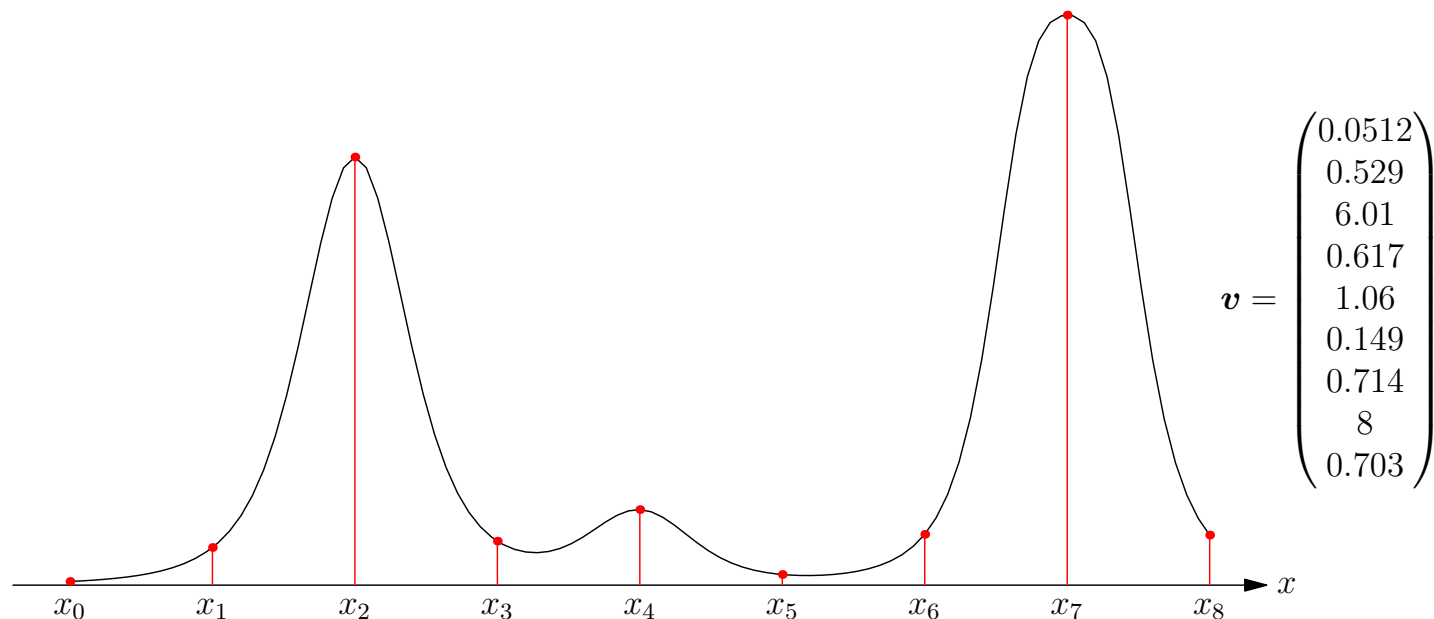
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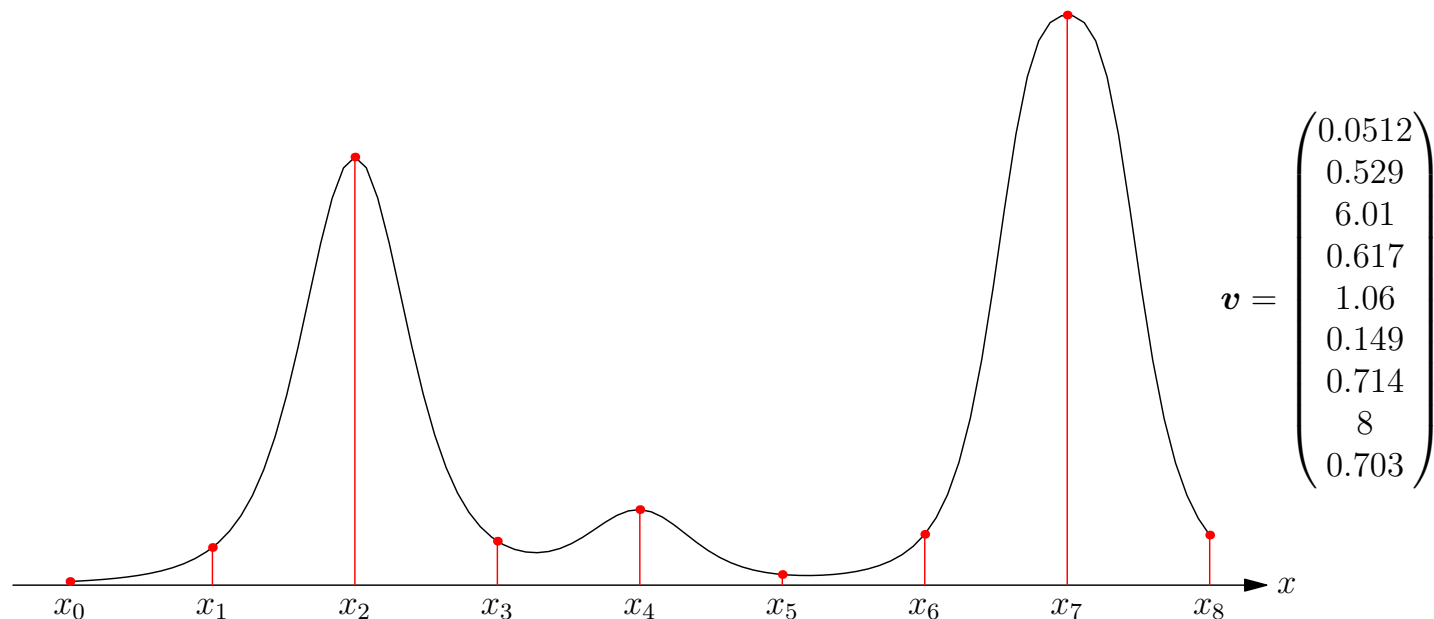
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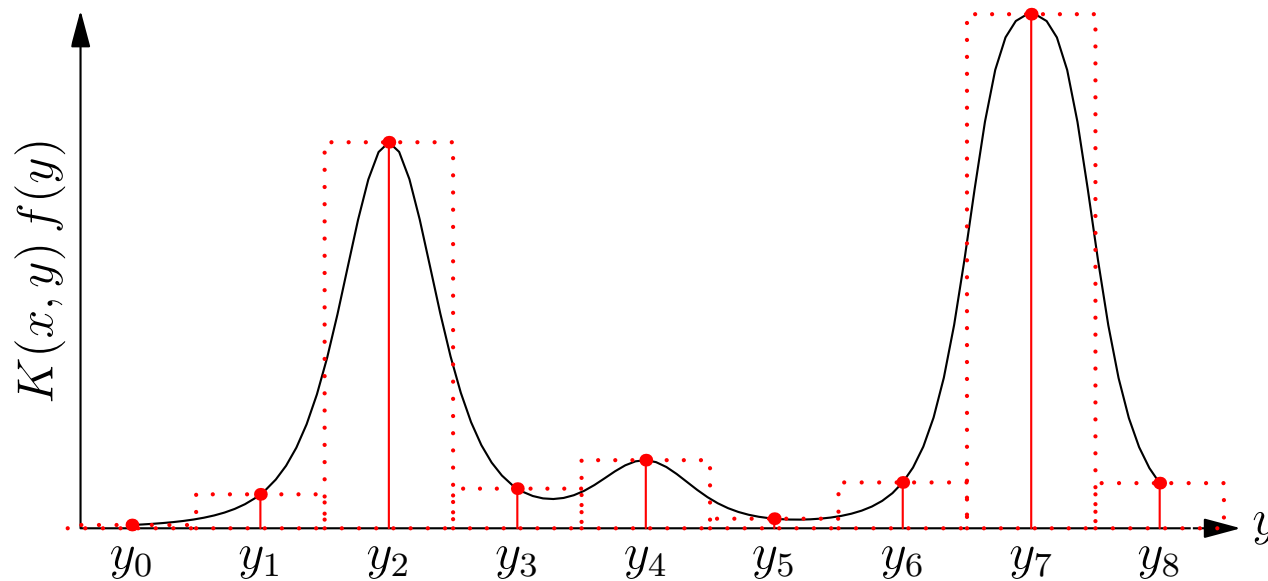


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Linear Operators

- Recall a linear function $\mathcal{T}[f(x)]$ can be represented by a kernel

$$\mathcal{T}[f(x)] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy$$

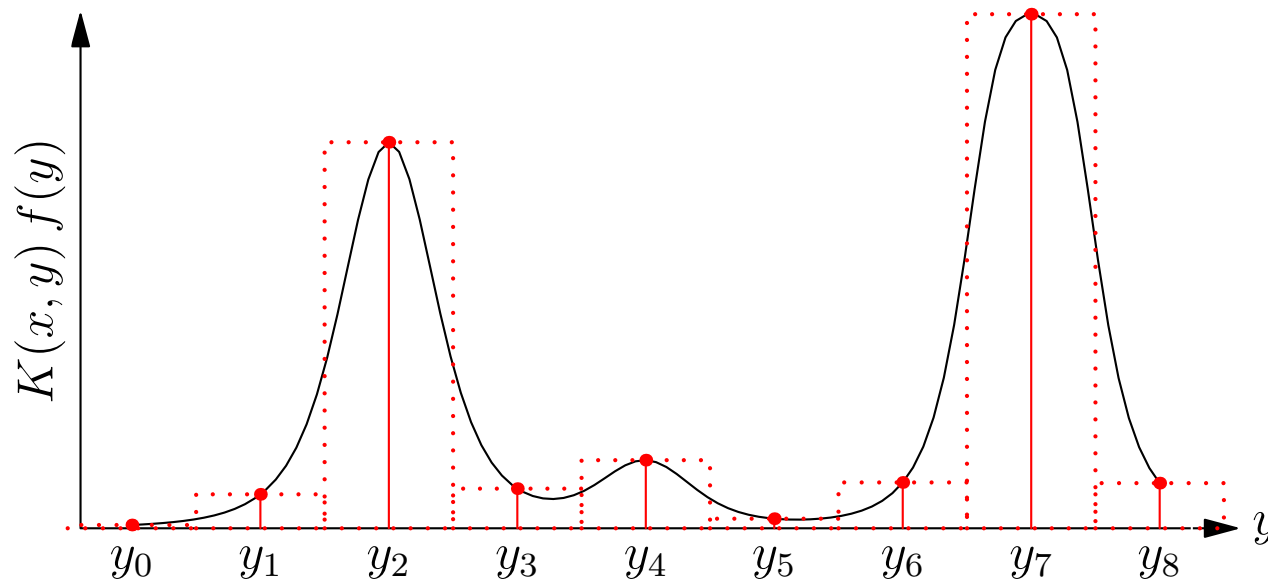


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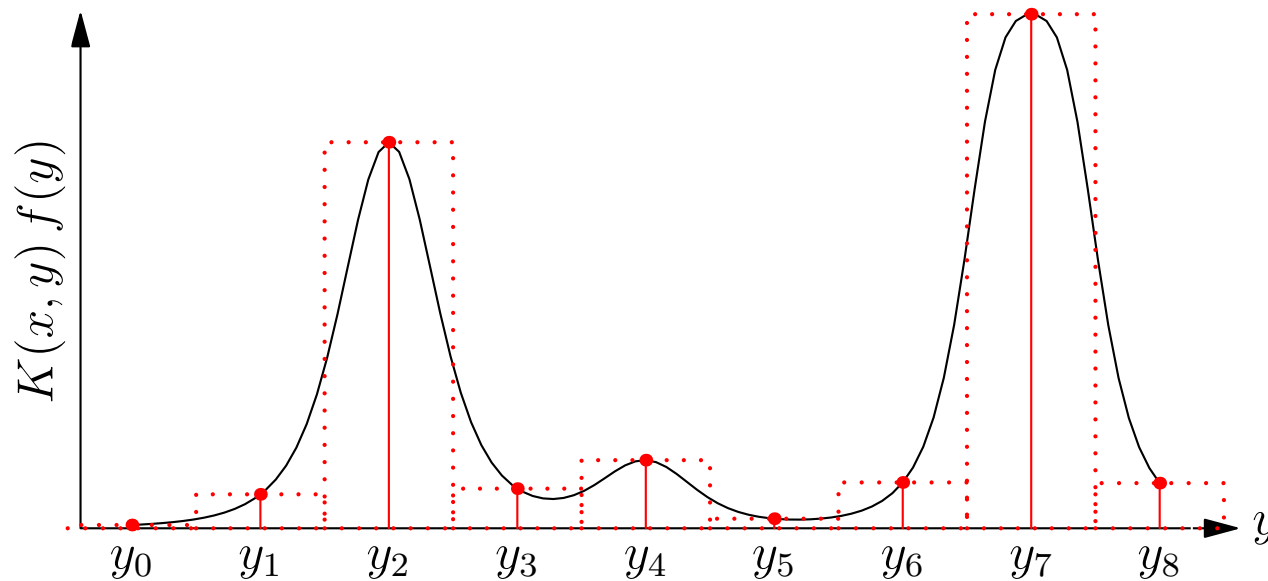


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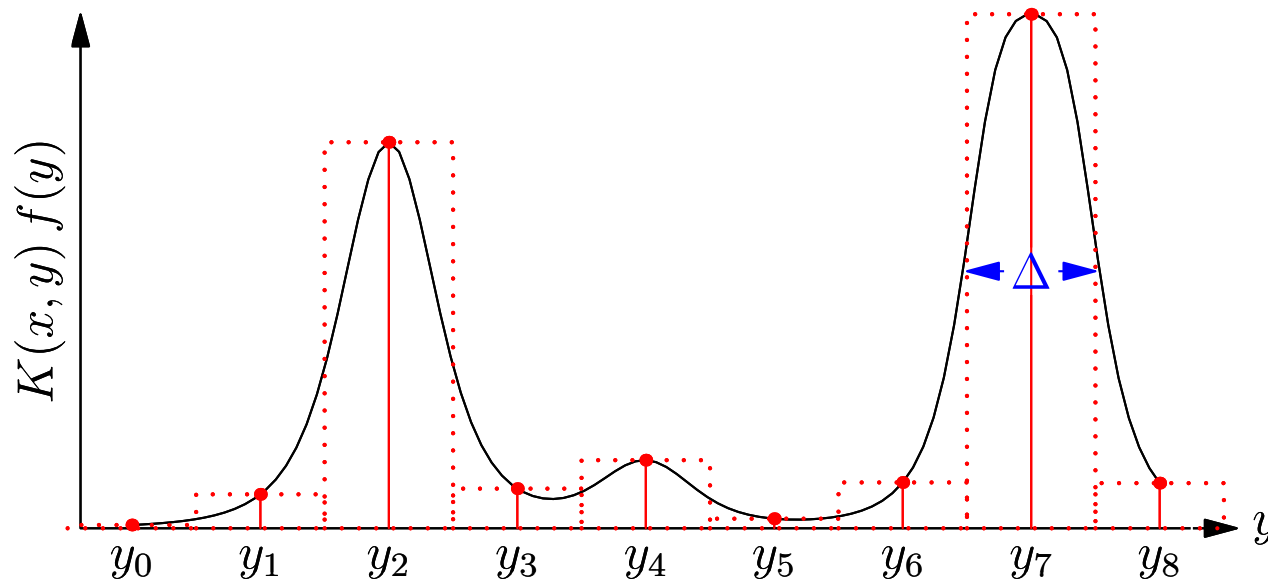


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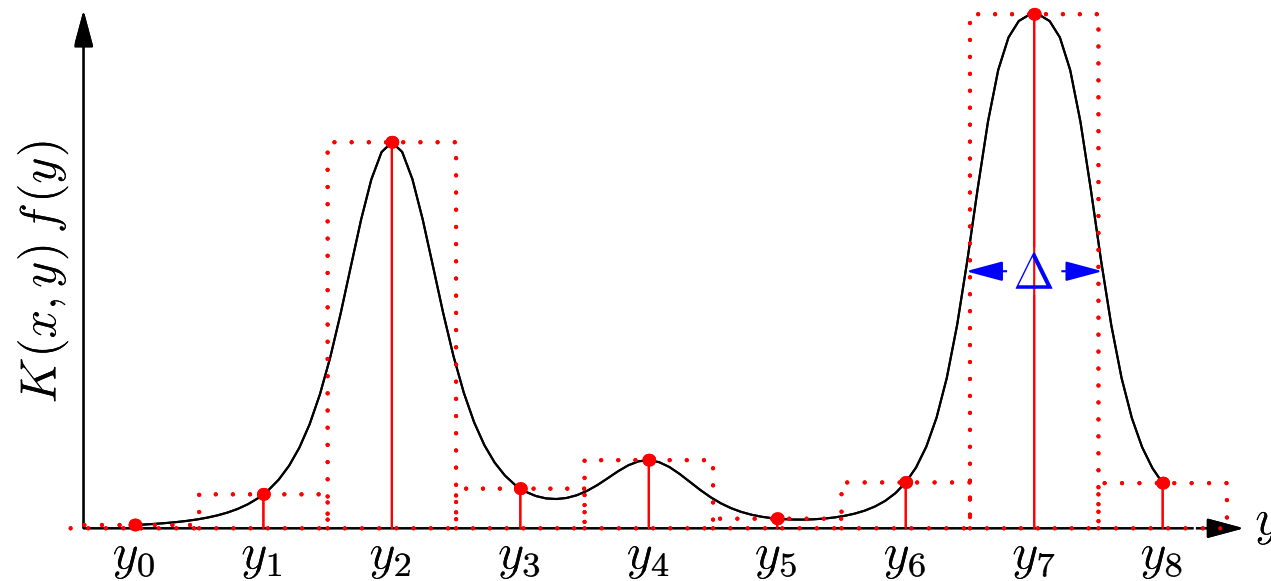


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- This is the definition of a SVM kernel we started with
- Note that for $\phi^{(k)}(\mathbf{x})$ to be real $\lambda^{(k)} \geq 0$ for all k
- If $\lambda^{(k)} < 0$ then $\langle \phi(\mathbf{x}), \phi(\mathbf{x}) \rangle = \|\phi(\mathbf{x})\|^2$ might be negative and “distance” between points in the extended feature space can be negative!
- If we use a kernel that isn’t positive semi-definite then the Hessian of the dual objective function will not be negative semi-definite and there will be a maximum where α diverges

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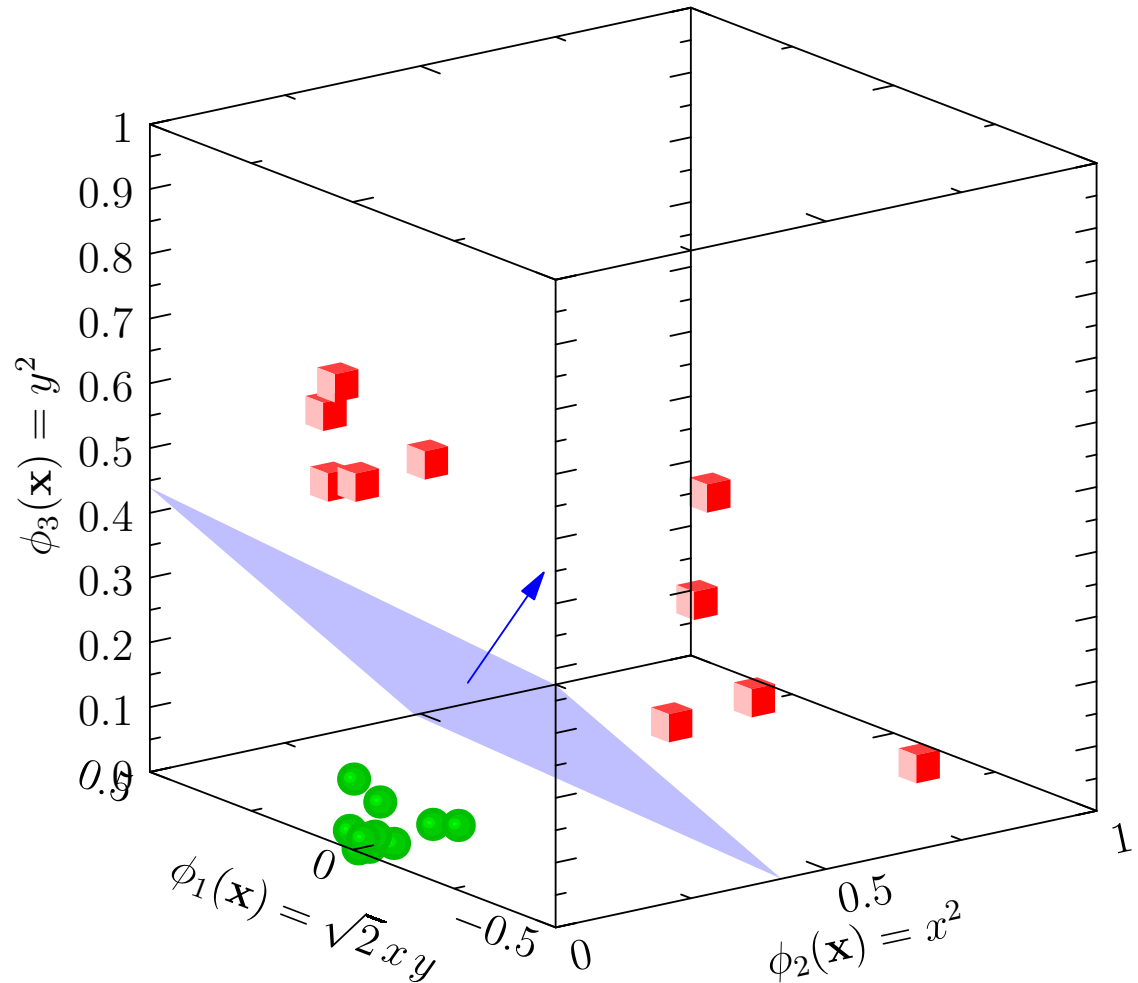
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- Note that for $\phi^{(k)}(\mathbf{x})$ to be real $\lambda^{(k)} \geq 0$ for all k
- If $\lambda^{(k)} < 0$ then $\langle \phi(\mathbf{x}), \phi(\mathbf{x}) \rangle = \|\phi(\mathbf{x})\|^2$ might be negative and “distance” between points in the extended feature space can be negative!
- If we use a kernel that isn't positive semi-definite then the Hessian of the dual objective function will not be negative semi-definite and there will be a maximum where α diverges

Outline

1. The Kernel Trick
2. Positive Semi-Definite Kernels
3. **Kernel Properties**
4. Beyond Classification



Positive Semi-Definite Kernels

- Kernels (or matrices) that have eigenvalues $\lambda^{(k)} \geq 0$ are called positive semi-definite
- (If the eigenvalues are strictly positive $\lambda^{(k)} > 0$ the kernels or matrices are called positive definite)
- Positive semi-definite kernels can always be decomposed into a sum of real functions

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Properties of Positive Semi-Definiteness

- Since

$$K(\boldsymbol{x}, \boldsymbol{y}) = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{y}) \rangle$$

- An immediate consequence is that for any function $f(\boldsymbol{x})$

$$\int f(\boldsymbol{x}) K(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) d\boldsymbol{x} d\boldsymbol{y} = \int f(\boldsymbol{x}) \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{y}) \rangle f(\boldsymbol{y}) d\boldsymbol{x} d\boldsymbol{y}$$

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Positive Semi-Definiteness

- The following statements are equivalent
 - ★ $K(\mathbf{x}, \mathbf{y})$ is positive semi-definite (written $K(\mathbf{x}, \mathbf{y}) \succeq 0$)
 - ★ The eigenvalues of $K(\mathbf{x}, \mathbf{y})$ are non-negative
 - ★ The kernel can be written

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Adding Kernels

- We can construct SVM kernels from other kernels
- If $K_1(\mathbf{x}, \mathbf{y})$ and $K_2(\mathbf{x}, \mathbf{y})$ are valid kernels then so is $K_3(\mathbf{x}, \mathbf{y}) = K_1(\mathbf{x}, \mathbf{y}) + K_2(\mathbf{x}, \mathbf{y})$

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Exponentiating Kernels

- If $K(\mathbf{x}, \mathbf{y})$ is a valid kernel so is $K^n(\mathbf{x}, \mathbf{y})$ (by induction)
 - ★ Assume $K(\mathbf{x}, \mathbf{y}) \succeq 0$ this satisfies base case
 - ★ If $K^{n-1}(\mathbf{x}, \mathbf{y}) \succeq 0$ then

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RBF Kernel

- Now $\mathbf{x}^\top \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$ is a valid kernel because it is an inner product of functions $\phi(\mathbf{x}) = \mathbf{x}$
- For $\gamma > 0$ we have $2\gamma \mathbf{x}^\top \mathbf{y} \succeq 0$
- Thus $\exp(2\gamma \mathbf{x}^\top \mathbf{y}) \succeq 0$
- If $K(\mathbf{x}, \mathbf{y}) \succeq 0$ then $g(\mathbf{x})K(\mathbf{x}, \mathbf{y})g(\mathbf{y}) \succeq 0$

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- If $K(\mathbf{x}, \mathbf{y}) \succeq 0$ then $g(\mathbf{x})K(\mathbf{x}, \mathbf{y})g(\mathbf{y}) \succeq 0$

$$\int f(\mathbf{x})g(\mathbf{x})K(\mathbf{x}, \mathbf{y})g(\mathbf{y})f(\mathbf{y})d\mathbf{x}d\mathbf{y} = \int h(\mathbf{x})K(\mathbf{x}, \mathbf{y})h(\mathbf{y})d\mathbf{x}d\mathbf{y} \geq 0$$

where $f(\mathbf{x})g(\mathbf{x}) = h(\mathbf{x})$

$$e^{-\gamma \mathbf{x}^\top \mathbf{x}} e^{2\gamma \mathbf{x}^\top \mathbf{y}} e^{-\gamma \mathbf{y}^\top \mathbf{y}} = e^{-\gamma \|\mathbf{x} - \mathbf{y}\|^2} \succeq 0$$

Other Kernels

- The success of SVMs has meant that researchers try to increase the area of application
- The condition that a SVM kernel must be positive semi-definite is quite restrictive
- There has been a cottage industry of researchers finding smart kernels for solving complicated problems
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String Kernels

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- This requires comparing strings
- There are a large number of kernels developed to do this

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Spectrum Kernel

- A simple way to compare documents is to collect a histogram of all occurrences of substrings of length p
- This is known as a p -spectrum
- A p -spectrum kernel counts the number of common substrings

$s = \text{statistics}$ $\mathcal{S}_3(s) = \{\text{sta}, \text{tat}, \text{ati}, \text{tis}, \text{ist}, \text{sti}, \text{tic}, \text{ics}\}$

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- A more sophisticated kernel is to count all of the common subsequences that occur in two documents
- Naively this would take an exponential amount of time to compute
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- String kernels for comparing subsequences are used in bioinformatics
- Kernels have been developed for comparing trees (e.g. for computer program evaluation, XML, etc.)
- Kernels have also been developed for comparing graphs (e.g. for comparing chemicals based on their molecular graph)

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- In an attempt to build kernels that capture more domain knowledge, kernels are constructed from other learning machines
- An example of this are “Fisher kernels” whose features come from an Hidden Markov Model (HMM) trained on the data
- These tend to have better discriminative power than the underlying model (HMM), and has a better feature set than a SVM using a generic kernel

Fisher Kernels

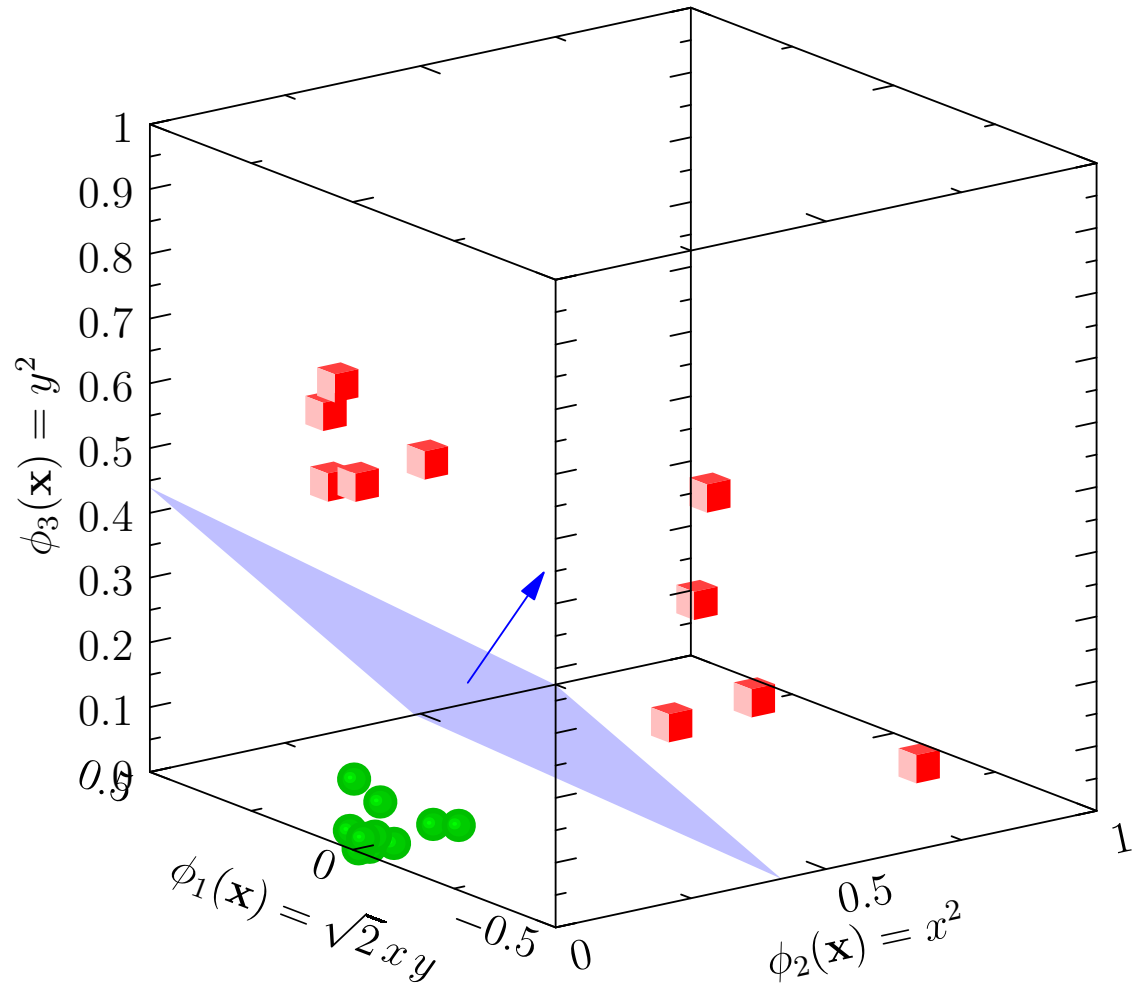
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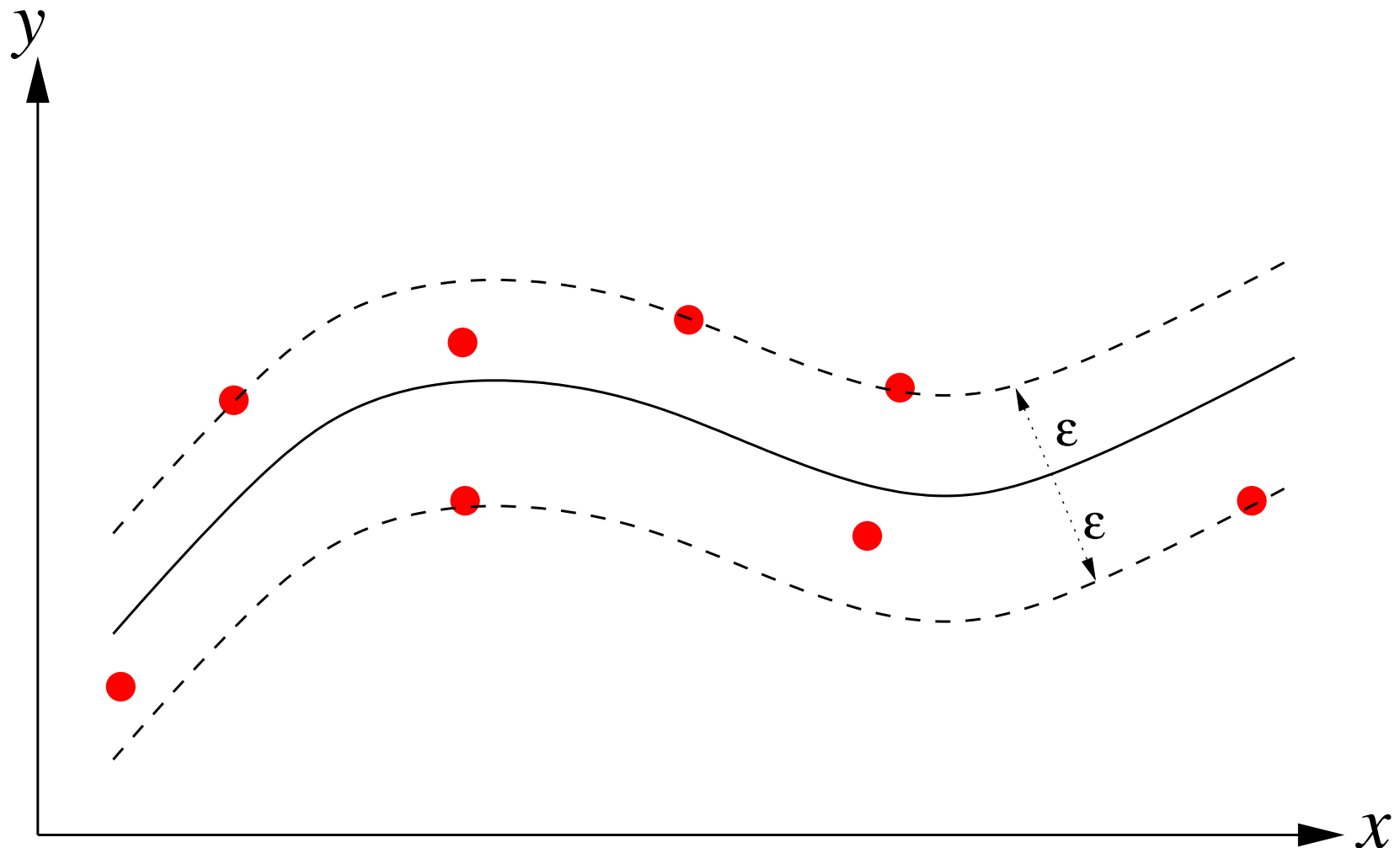
Outline

1. The Kernel Trick
2. Positive Semi-Definite Kernels
3. Kernel Properties
4. **Beyond Classification**



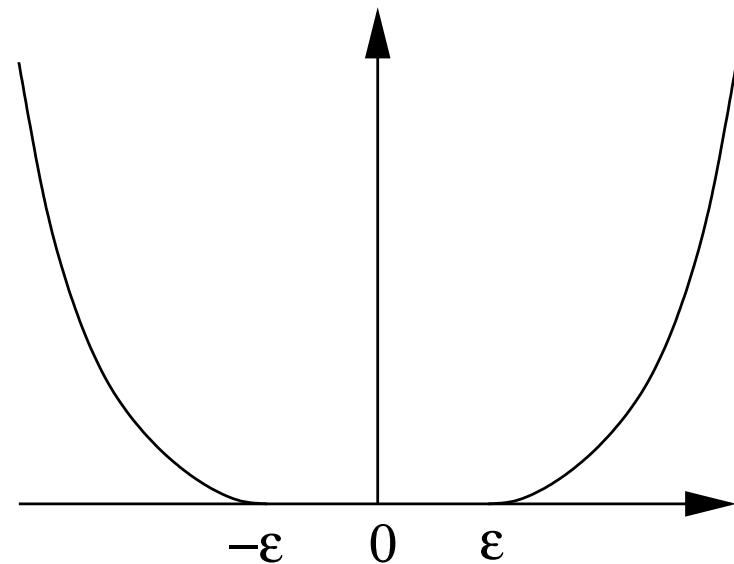
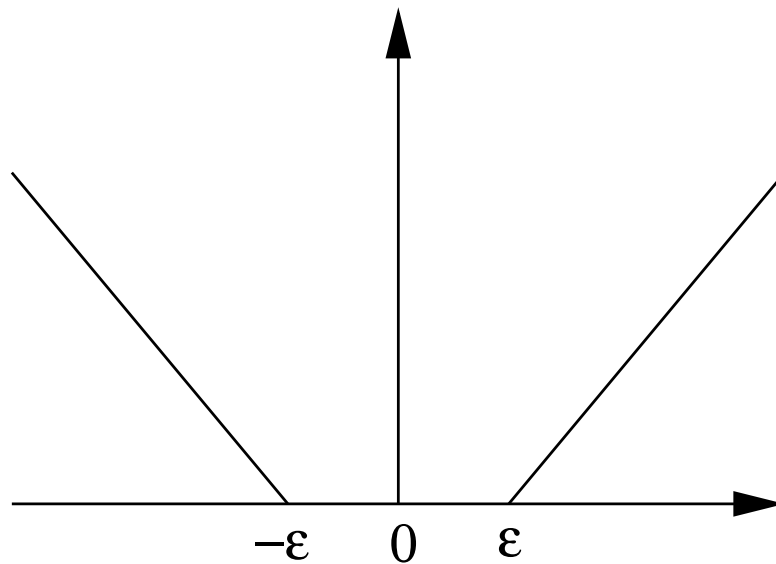
Regression with Margins

- SVMs can be modified to perform regression



Error Functions

- Can introduce slack variables with different errors



- This can be transformed to a quadratic programming problem

Ridge Regression Using Kernels

- We can also solve regression problems without using margins
- To solve a regression problem once again the problem is set up as a quadratic programming problem

$$\min_{\mathbf{w}} \lambda \|\mathbf{w}\|^2 + \sum_{i=1}^m (y_i - \mathbf{w}^\top \phi(\mathbf{x}_i))^2$$

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Kernel Methods

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 - ★ Kernel principle component analysis (KPCA)
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